Raspberry Pi (mounted face-down) + HAT ID

Fájl: pi.kicad_sch

Audio: 2x TAS5805M (stereo BTL + sub PBTL)

Fájl: audio.kicad_sch

Power: USB-C PD charger (IP2366), 5V/5A Pi buck, sub boost, 3V3 LDOs

Fájl: power.kicad_sch

Battery: 3S 18650 clips, BMS module, protection

Fájl: battery.kicad_sch

UI + supervisor MCU (CH32V003), OLED, buttons, encoder, USB console

Fájl: ui.kicad_sch

Z2Labs – Bassberry (open hardware, CERN-OHL-S-2.0)

Sheet: /

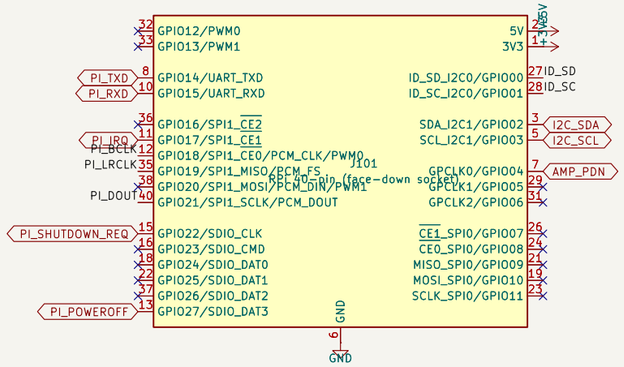
File: bassberry.kicad_sch

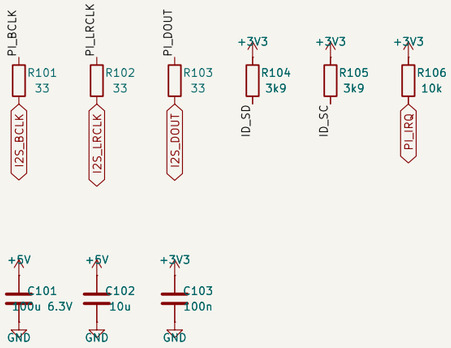
Title: Bassberry 2.1 – top level

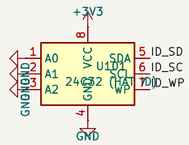
Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/6

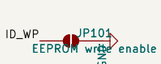
Raspberry Pi plugs in FACE-DOWN: footprint is the mirrored face-down socket.

I2C1 has 1k8 pull-ups on the Pi itself. Series resistors = source termination for I2S.











Z2Labs – Bassberry (open hardware, CERN-OHL-S-2.0)

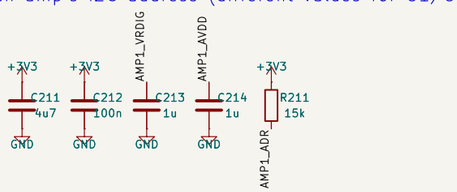
Sheet: /Raspberry Pi (mounted face-down) + HAT ID/
File: pi.kicad_sch

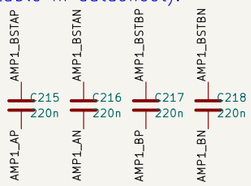
Title: Raspberry Pi (mounted face-down) + HAT ID

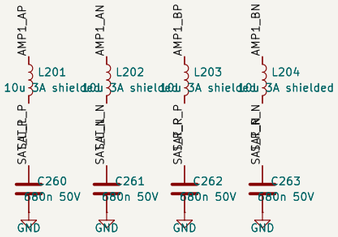
Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/6

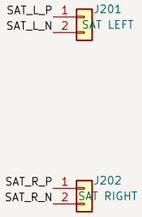
U2 sub amp: PBTL. OUT_A+/OUT_A− and OUT_B+/OUT_B− paralleled pre−filter (verify vs TI datasheet PBTL section).

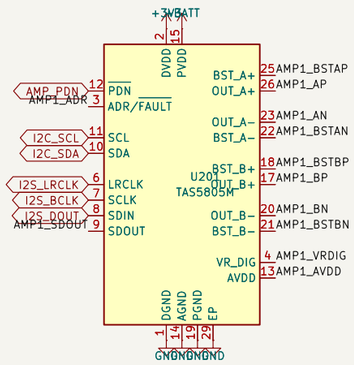
ADR resistor sets each amp's I2C address (different values for U1/U2, verify table in datasheet).

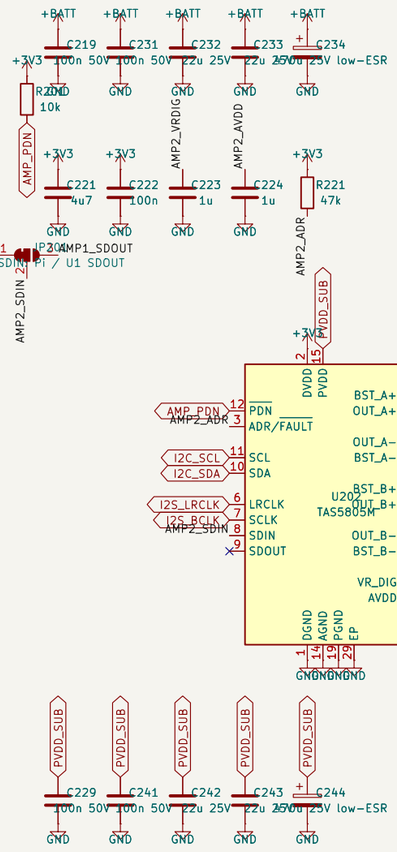


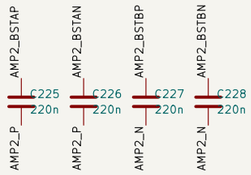


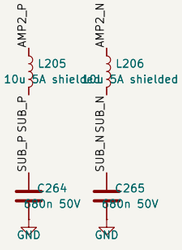


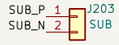












Z2Labs − Bassberry (open hardware, CERN−OHL−S−2.0)

Sheet: /Audio: 2x TAS5805M (stereo BTL + sub PBTL)/
File: audio.kicad_sch

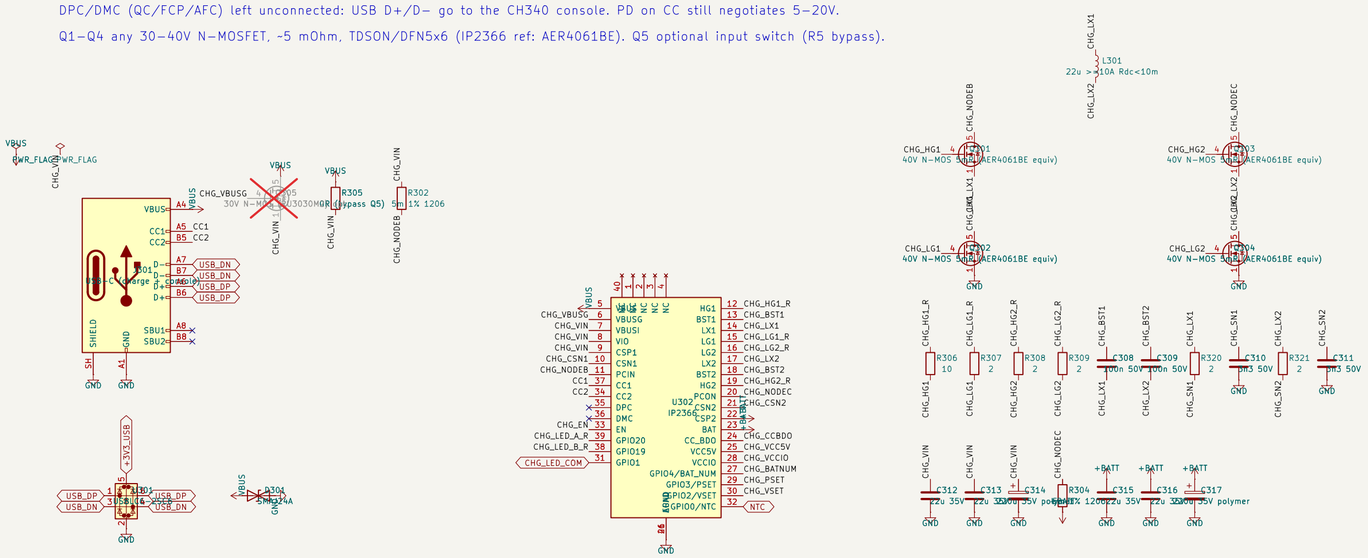
Title: Audio: 2x TAS5805M (stereo BTL + sub PBTL)

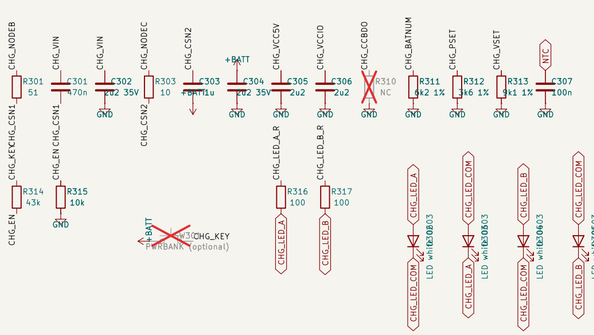
Size: A3	Date: 2026−10−26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/6

IP2366 application per Injoinic datasheet V1.10 fig. 9. Settings: 3S (6.2k), 4.2V/cell (9.1k), 30W (3.6k).

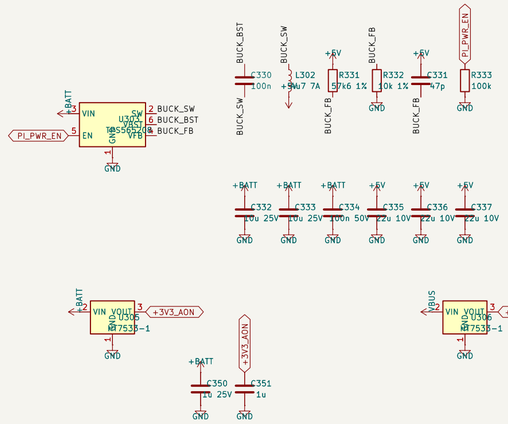
DPC/DMC (QC/FCP/AFC) left unconnected: USB D+/D− go to the CH340 console. PD on CC still negotiates 5–20V.

Q1–Q4 any 30–40V N-MOSFET, ~5 mOhm, TDSON/DFN5x6 (IP2366 ref: AER4061BE). Q5 optional input switch (R5 bypass).

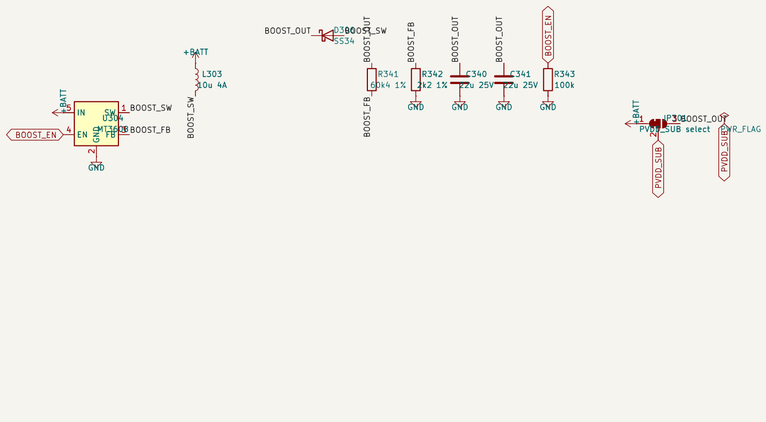




Pi 5V buck: Vout = 0.76*(1+R31/R32) = 5.14 V. EN driven by supervisor (PI_PWR_EN).



Sub PVDD boost (optional): Vout = 0.6*(1+R41/R42) = 17.1 V. JP1 selects PVDD_SUB = +BATT (default) or boost.



Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)
Sheet: /Power: USB-C PD charger (IP2366), 5V/5A Pi buck, sub boost, 3V3 LDOs/
File: power.kicad_sch

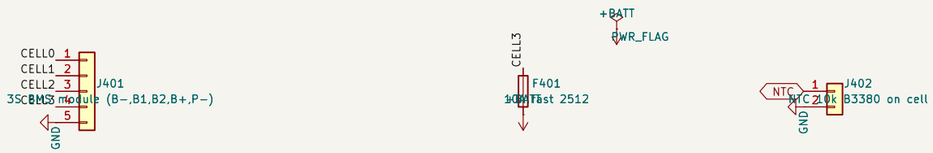
Title: Power: USB-C PD charger (IP2366), 5V/5A Pi buck, sub boost, 3V3 LD

Size: A2	Date: 2026-10-26	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 4/6

3x 18650 in Keystone 54 clips. Balancing + protection by an off-the-shelf 3S BMS module (B-, B1, B2, B+, P-).

Reverse-cell LED lights only when a cell is inserted backwards. P- of the BMS = board GND.





Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)

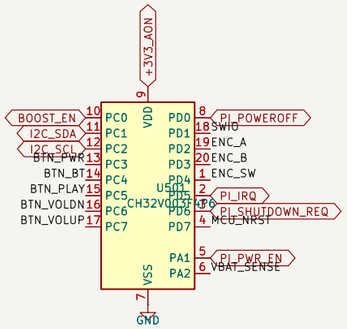
Sheet: /Battery: 3S 18650 clips, BMS module, protection/
File: battery.kicad_sch

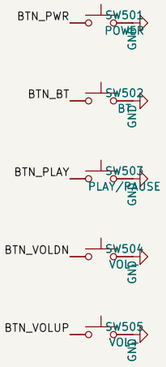
Title: Battery: 3S 18650 clips, BMS module, protection

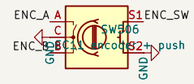
Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/6

Supervisor runs from +3V3_AON. I2C slave to the Pi on PC1/PC2; PI_IRQ and PI_SHUTDOWN_REQ are open-drain.

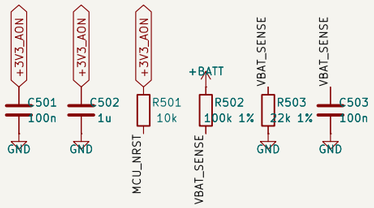
Buttons active-low with MCU internal pull-ups. VBAT_SENSE = VBAT * 22/122.

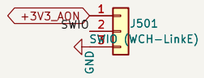


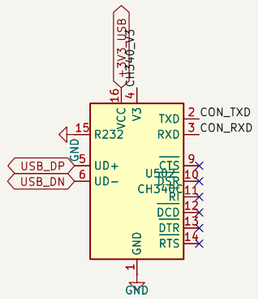


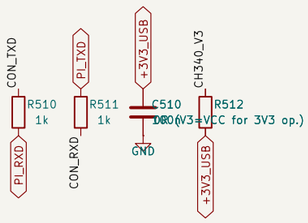












Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)

Sheet: /UI + supervisor MCU (CH32V003), OLED, buttons, encoder, USB console/

File: ui.kicad_sch

Title: UI + supervisor MCU (CH32V003), OLED, buttons, encoder, USB console

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/6